

Bernardo Bahia Monteiro

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Experience

Graduate Student Research Assistant

Active Aeroelasticity and Structures Research Laboratory (A²SRL)

University of Michigan
September 2019–present

Research interest: Incorporate closed-loop control considerations in the Multidisciplinary Design Optimization (MDO) of flexible aircraft

Responsibilities: Maintain and improve in-house flexible aircraft simulation software (C++/Python)
Program real time embedded systems for wind tunnel testing (C/C++)
Operate aero-servo-elastic model during wind tunnel tests
Design and analysis of aero-servo-elastic wind tunnel model
Administrate the lab's Linux-based high performance computers

Intern at Embraer

Flight Operations Engineering

São José dos Campos, Brazil
January, 2018–January, 2019

Design and implement software solutions to meet costumers demands (Java/C)
Support meetings with airline customers
Develop proof-of-concept data analysis solutions (Python)
Worked in an agile development team

Preliminary Design

February, 2019– June, 2019

Program semi-empirical aerodynamic models for the in-house MDAO code (Fortran)

Voluntary Research Assistant at UFMG

Belo Horizonte, Brazil
2015–2018

Worked with Prof. Ricardo Luiz Utsch de Freitas Pinto

Numerical Analysis

Preprint: *End-point corrections for the midpoint rule*; available at [arXiv:1812.00243](https://arxiv.org/abs/1812.00243)

Voluntary Summer Research Assistant at Strathclyde

Glasgow, UK
2017

Project *Future UK Small Payload Launcher*

Feasibility study of a horizontal rocket launcher with a reusable first stage

Worked with Multidisciplinary Design Optimization and Optimal Control

Model Rocketry Team at UFMG

Belo Horizonte, Brazil
2014–2016

Designed and manufactured two model rockets for a nationwide competition

Coordinated a team of 8 people responsible for the project of one of these rockets

Responsible for rocket stability

Education

University of Michigan

PhD Aerospace Engineering

Ann Arbor, MI
September 2019–May 2024 (expected)

GPA: 4.00

Advisors: Carlos Cesnik, Ilya Kolmanovsky

Selected courses: MDO, Aeroelasticity, Structural Dynamics, Applied Nonlinear Dynamics,
Linear Feedback Control, Adaptive Control

Universidade Federal de Minas Gerais

BSc Aerospace Engineering

Belo Horizonte, Brazil
February 2014–June 2019

GPA: 4.9/5.0

Admitted in first place, January 2014

Graduated first in class

University of Strathclyde

One semester exchange in Aero-mechanical Engineering

First class result in all disciplines, mostly at master's level: Control Systems Design, Introduction to Engineering Optimization, Spaceflight Systems, Aerospace Propulsion, Engineering Composites, Engineering Analysis 3 (Introduction to CFD).
Development of a dynamical inversion control system for a drone (coursework)

Glasgow, UK
January–July, 2017

Short term courses

Embraer's

Flight Operations Engineering Course

Workload: 30 hours

São José dos Campos, Brazil
April 9–12, 2018

Embraer's

Basic and Advanced Fortran

Workload: 40 hours

São José dos Campos, Brazil
April 2–6, 2018

Brazilian — German Summer School on

Numerical Simulations with the Finite Element Method

Mathematical Analysis and Computational Practice

Lecturers: Prof. Dr. Thomas Richter (Erlangen-Nürnberg University)

Prof. Dr. Malte Braack (Kiel University)

Workload: 30 theoretical and 25 practical classes (of 45min each)

UFRGS, Porto Alegre, Brazil
February 15–26, 2016

Awards and achievements

- 2nd overall place on ENEM 2013 (National High School Exam), more than 7 million students applied
- Gold Medal on the XV Brazilian Olympiad of Astronomy and Astronautics (October 2012)
- Honorable Mention on the state's Olympiad of Mathematics (MG, 2007)

Computer skills

- MATLAB/SIMULINK
- C/C++, Python, Java, Fortran
- Linux/Unix
- Git
- L^AT_EX

Linguistic skills

- Proficient English (Level C2 — TOEFL: 117 / CAE Grade A)
- Native Portuguese

Hobbies

Mountain biking, tennis, gliding, road cycling, skiing, motorcycle/bicycle maintenance